MD ASHIKUL HAQUE

mdashikul.haque@utdallas.edu | +1 (515) 598 6458 | ashikul-haque.github.io
Google Scholar: scholar.google.com/citations?user=11vTHPYAAAAJ

EDUCATION

PhD, Computer Science Expected May 2027
University of Texas at Dallas, USA
Thesis: *Resilient and Secure LPWAN Communication Under Jamming and Coexistence*
Advisor: Prof. Abusayeed Saifullah

MSc, Computer Science 2024
Wayne State University, USA

BSc, Computer Science and Engineering 2018
Bangladesh University of Engineering and Technology, Bangladesh

RESEARCH INTERESTS

Real-Time Scheduling for LLM Agents • Cross-Technology Communication • Resilient and Secure Communication • Learning-Based Coexistence Management in LPWANs • Real-Time Scheduling for Industrial Networks

Research Summary. I build wireless and cyber-physical systems that stay correct and on time when the resource they depend on is contested, whether that resource is a token budget, airtime, spectrum, or the energy left in a battery. My work covers deadline-aware token scheduling for large language model (LLM) agents that drive physical actuators, cross-technology communication from Long Range Frequency Hopping Spread Spectrum (LR-FHSS) to LoRa, and physical-layer defenses that keep LoRa networks decoding under jamming attacks. I also work on learning-based coexistence management for dense low-power wide-area network (LPWAN) deployments, and on medium access control (MAC) and scheduling mechanisms for bursty and mixed-criticality traffic in industrial wireless networks. I validate these designs through PyTorch-based learning, NS-3 simulation, SDR prototyping with GNU Radio and USRP B200, and outdoor and testbed experiments.

CONFERENCE PUBLICATIONS

[9] Md Ashikul Haque, Abusayeed Saifullah. *Deadline-Aware Token Scheduling for Physical-I/O LLM Agents*. In Proc. IEEE Real-Time Systems Symposium (RTSS 2026), pp. 1–12. Acceptance rate: 14%.

[8] Md Ashikul Haque, Aakriti Jain, Venkata Prashant Modekurthy, Abusayeed Saifullah. *Enabling Cross Technology Communication from LR-FHSS to LoRa*. In Proc. ACM/IEEE International Conference on Embedded Networked Sensor Systems (SenSys 2026), CPS-IoT Week 2026. Acceptance rate: 19.8%.

[7] Md Ashikul Haque, Abusayeed Saifullah. *Decoding LoRa Packets under Collaborative Jamming Attacks*. In Proc. 23rd International Conference on Embedded Wireless Systems and Networks (EWSN 2026). Acceptance rate: 23.5%.

[6] Md Ashikul Haque, Abusayeed Saifullah. *ReMix: Resource-Reclaiming Mixed-Criticality Scheduling for Industrial IoT*. In Proc. The 35th International Conference on Computer Communications and Networks (ICCCN 2026), pp. 1–10. Acceptance rate: 25%.

[5] Md Ashikul Haque, Abusayeed Saifullah. *Mitigating Jamming Attacks in LoRa Networks: A Defense Strategy Against LoRa Based Jammers*. In Proc. ACM International Symposium on Mobile Ad Hoc Networking and Computing (MobiHoc 2025). Acceptance rate: 23%.

[4] Md Ashikul Haque, Abusayeed Saifullah, Haibo Zhang. *Deep Reinforcement Learning Based Coexistence Management in LPWAN*. In Proc. IEEE Conference on Computer Communications (INFOCOM 2025). Acceptance rate: 18.6%.

[3] Md Ashikul Haque, Abusayeed Saifullah. *Handling Jamming Attacks in LoRa Networks*. In Proc. ACM/IEEE Conference on Internet of Things Design and Implementation (IoTDI 2024), CPS-IoT Week 2024.

[2] Aakriti Jain, Md Ashikul Haque, Abusayeed Saifullah, Haibo Zhang. *Burst-MAC: A MAC Protocol for Handling Burst Traffic in LoRa Network*. In Proc. IEEE Real-Time Systems Symposium (RTSS 2024).

[1] Md Ashikul Haque, Abusayeed Saifullah. *A Game-Theoretic Approach for Mitigating Jamming Attacks in LPWAN*. In Proc. International Conference on Embedded Wireless Systems and Networks (EWSN 2023).

WORKSHOP PUBLICATIONS

[1] Md Ashikul Haque, Abusayeed Saifullah, Haibo Zhang. *Coordinating Coexisting Learning Agents in Shared Spectrum via Parameter Space Complementarity*. Agents in the Wild: Safety, Security, and Beyond (ICLR 2026).

[2] Md Ashikul Haque, Dali Ismail, Abusayeed Saifullah. *Scalable Real-Time Control in Industrial Cyber-Physical Systems*. In Proc. Workshop on Software-Defined Networking for Cyber-Physical System Automation (SDN-CPS 2024), co-located with ICDCN 2024, pp. 353–358.

MANUSCRIPTS UNDER REVIEW

Md Ashikul Haque, Abusayeed Saifullah, Haibo Zhang.
Generative Policy Coordination for Coexisting Learning-Enabled LPWANs.

SELECTED TALKS

ReMix: Resource-Reclaiming Mixed-Criticality Scheduling for Industrial IoT — Jul 2026
IEEE ICCCN 2026, Honolulu, Hawaii, USA

Mitigating Jamming Attacks in LoRa Networks: A Defense Strategy against LoRa Based Jammers — Oct 2025
ACM MobiHoc 2025, Rice University, Houston, USA

Handling Jamming Attacks in a LoRa Network — May 2024
ACM/IEEE IoTDI 2024, Hong Kong, China

RESEARCH AND PROFESSIONAL EXPERIENCE

Graduate Research Assistant, Dept. of Computer Science — 2022 to Present
The University of Texas at Dallas, USA | Advisor: Prof. Abusayeed Saifullah

- Designed and implemented cross-technology communication mechanisms from LR-FHSS to LoRa using GNU Radio and USRP, enabling long-range downlink and control for LPWAN deployments.
- Developed signal-processing and protocol mechanisms to improve LPWAN resilience against reactive and collaborative jamming, with evaluation on real testbeds and outdoor experiments.
- Built learning-based coexistence management and telemetry-driven evaluation pipelines for LPWANs using PyTorch, NS-3, and measurement-driven experimentation.

Assistant Engineer, ICT Division — 2020 to 2021
Dhaka Electric Supply Company, Bangladesh

- Maintained ICT infrastructure with monitoring, incident response, and reliability focused operations for distributed utility systems.

Software Engineer, IoT Division — 2019 to 2020
Samsung R&D Institute Bangladesh, Bangladesh

- Developed production grade C++ components for Samsung Cloud client integration supporting SmartThings workflows and IoT device communication.
- Delivered iOS integration via Swift wrapper and onboarding features, collaborating across client libraries and application layers.
- Automated multi step build and packaging workflows using Bash to improve reproducibility and reduce manual release effort.

TEACHING

Graduate Teaching Assistant, Dept. of Computer Science — 2026
The University of Texas at Dallas, USA
Courses: Computer Networks; Performance of Computer Systems and Networks

- Graded assignments, exams, and project deliverables for Computer Networks and Performance of Computer Systems and Networks.
- Helped students understand course concepts through office hours, question answering, and discussion of networking and systems topics.
- Supported course projects by clarifying requirements, troubleshooting implementations, and providing feedback on design and debugging.

Guest Lecturer (multiple sessions) — 2025

The University of Texas at Dallas, USA
Course: Internet of Things

Lecturer, Dept. of Computer Science 2018 to 2019
Millennium University, Bangladesh
Courses: Structured Programming; Data Communication

Awards and Grants

Dean's List Award, Wayne State University 2024
Professional Travel Award, College of Engineering, Wayne State University 2024
SIGBED Travel Grant, CPS-IoT Week, San Antonio, Texas, USA 2023

Professional Service

Conference Referee
- ACM/IEEE Intl. Conf. on Embedded Networked Sensor Systems (SenSys) 2022, 2024, 2025, 2026
- IEEE Real-Time Systems Symposium (RTSS) 2023, 2024, 2025
- IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS) 2023, 2024
- Intl. Conf. on Embedded Wireless Systems and Networks (EWSN) 2023, 2024, 2025
- IEEE Intl. Conf. on Distributed Computing in Smart Systems and IoT (DCOSS) 2024

Conference Reviewer
- IEEE Real-Time Computing Systems and Applications (RTCSA) 2024, 2025
- IEEE Intl. Conf. on Sensing, Communication, and Networking (SECON) 2026
- IEEE Intl. Conf. on Distributed Computing in Smart Systems and IoT (DCOSS) 2026

Journal Referee
- IEEE/ACM Transactions on Networking 2024, 2025

Journal Reviewer
- Pervasive and Mobile Computing (Elsevier) 2023

Volunteer: CPS-IoT Week 2023

Technical Skills

Programming: C, C++, Python, MATLAB, SQL, Swift, Bash
ML and Data: PyTorch, Pandas
Wireless: GNU Radio, SDR pipelines, Telemetry
Simulation: NS-3, Linux, Docker, Git, LaTeX
Hardware: USRP B200, LR-FHSS, LoRa and LPWAN devices, IoT gateways

References

Dr. Abusayeed Saifullah
Associate Professor, Dept. of Computer Science
University of Texas at Dallas
abusayeed.saifullah@utdallas.edu
+1 (972) 883.4171

Dr. Haibo Zhang
Associate Professor, School of Computer Science
and Engineering, UNSW Sydney
haibo.zhang@unsw.edu.au
+61 2 9385 1000

Dr. Prashant Modekurthy
Assistant Professor, Dept. of Computer Science
University of Nevada, Las Vegas (UNLV)
prashant.modekurthy@unlv.edu
+1 (702) 774.3405